

Analysis First-Year Exam, January 2017, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $f_n : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable and let f'_n converge to g uniformly on $[a, b]$. Prove that if f_n converges pointwise at a then f_n converges uniformly on $[a, b]$ to a continuously differentiable function f such that $f' = g$.
2. Let f be a real-valued function on $[a, b]$. For each of the following statements, give a proof or a counterexample, as appropriate.
 - (i) If f is Riemann integrable with strictly positive Riemann integral over $[a, b]$, then f is strictly positive on some nonempty open interval.
 - (ii) The same statement, with Riemann replaced by Lebesgue throughout.
3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. For $a \leq b$ and for N a natural number, consider the statement: “if $\int_a^b f(t)t^{2n+1} dt = 0$ for each integer $n \geq N$, then $f = 0$ on $[a, b]$.” Argue the truth or falsity of this statement in the following cases:
 - (i) $N = 0$, $[a, b] = [0, 1]$;
 - (ii) $N = 1$, $[a, b] = [0, 1]$;
 - (iii) $N = 0$, $[a, b] = [-1, 1]$.
4. Let f_n be a sequence of measurable real-valued functions on the measurable space (Ω, Σ) . Prove that each of the following sets is measurable:
 - (i) the set P comprising all $\omega \in \Omega$ such that $f_n(\omega)$ converges to an irrational number;
 - (ii) the set Q comprising all $\omega \in \Omega$ such that $f_n(\omega)$ converges to a rational number;
 - (iii) the set R comprising all $\omega \in \Omega$ such that $f_n(\omega)$ converges to a real number.
5. Let f be an integrable function on the measure space (Ω, Σ, μ) . Prove that for each $\varepsilon > 0$ there exists a $\delta > 0$ such that if $A \in \Sigma$ and $\mu(A) < \delta$, then $\int_A |f| d\mu < \varepsilon$.